Experimental Evaluation of Delayed-based Detectors against Power-off Attack

Maryam Esmaeilian*, Aghiles Douadi‡, Zahra Kazemi*, Vincent Beroulle*, Amir-Pasha Mirbaha*, Mahdi Fazeli†, Elena Ioana Vatajelu‡, Paolo Maistri‡, Giorgio Di Natale‡
*Univ. Grenoble Alpes, Grenoble INP, LCIS, 26000 Valence, France
†, School of Information Technology, Halmstad University, Halmstad, Sweden
‡Univ. Grenoble Alpes, CNRS, Grenoble INP1 , TIMA, 38000 Grenoble, France

2023 IEEE 29th International Symposium on On-Line Testing and Robust System Design (IOLTS) | 979-8-3503-4135-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/IOLTS59296.2023.10224876

***Abstract*—Embedded systems are vulnerable to significant security threats from Fault Injection Attacks (FIAs), which allow attackers to gain access to confidential information. While various attack detectors have been proposed in the literature to detect different types of FIAs, these detectors themselves are susceptible to such attacks and can be compromised. Hence, the robustness of these detectors is critical in maintaining the security of embedded systems. The focus of this study is to evaluate the robustness of digital circuits and delay-based digital detectors against a new type of FIA called Power-Off Attack (POA). POA occurs when the power to the chip is turned off, and the detectors are not active. Following a POA attack, the circuit or its detectors may not function properly when the power is turned back on, which can allow other attacks to be applied without being detected if the detectors are less sensitive. This study implements two detectors on Xilinx Artix-7 FPGAs and examines the impact of heating cycles on detector characteristics when the FPGA is in various states, including power-off, power-on, and inactive states (such as clock-freezing mode). Our experiments reveal that heating cycles in power-off mode can alter the FPGA component delays and the accuracy of its detectors, which highlights the vulnerability of these systems to POA and potential issues for embedded system security.**

***Index Terms*—hardware security, fault attack, power-off attack, temperature attack, secure circuit design, delay-based detectors**

I. Introduction

Nowadays, the use of electronic devices in our daily life is growing fast. These devices are employed in security applications, such as authentication applications. One of the most important concerns about such security applications is their existing vulnerabilities against different attacks. Among these attacks, Hardware Attacks (HAs) against embedded devices have attracted a lot of attention. Side Channel Attacks (SCA) [1] and Fault Injection Attacks (FIA) [2] are two types of these attacks. Our focus in this study is on the second type. FIAs involve the attacker attempting to manipulate the system's input or environmental conditions, such as temperature, to cause unintended behavior and extract confidential information. The techniques for performing Fault Injection Attacks (FIAs) are becoming increasingly more advanced and powerful. For instance, various digital and analog detectors have been proposed to detect the attack before it causes the wrong behavior in the devices [3] [4]. Recent works are mostly focused on digital detectors because they can be easily calibrated and placed close to security primitives such as PUFs and encryption cores. For example, in [5] a digital detector is presented to detect FIA based on electromagnetic radiation. Furthermore, digital detectors introduced in [6] and [7] can detect clock glitching and voltage glitching attacks, respectively. Therefore, detectors are used to protect the system from security attacks such as FIA. However, it is critical to ensure the protection of these detectors themselves against potential FIAs. There is a bunch of research work on the vulnerability of detectors [8] [9] to various types of Fault Injection Attacks (FIAs) as well as methods for protecting them. All of them assume that the detector is connected to the power supply, but none of them have been evaluated against FIA when the power is off. This new type of FIA, which we call a Power-Off Attack (POA), is performed when the target device is not connected to any power supply. The characteristics of the detectors can be changed by an attacker without being detected when the detectors are off. The aim of this study is to assess how environmental changes, specifically heating, impact the properties of simple digital detectors when power is not supplied. In order to conduct this experiment, we subjected the chip to varying periods of heating, as certain Fault Injection Attacks (FIAs) like Laser Fault Injection (LFI) can lead to temperature elevation. By applying heat, we were able to evaluate the impact of the FIA on the chip's properties and determine its effects on the detection threshold.

The rest of the paper is organized as follows: In Section II, the structure and methodology of this study will be explained. Experimental results are presented in Section III. In Section IV, we discuss the results. And at the end of this paper, in Section V, we conclude and give perspectives for future works.

II. Our Methodology

The goal of this study is to evaluate how high temperatures when the power is off affect the sensitivity of the detectors. To achieve this objective, we have selected two main detector designs for our experiment. We chose to evaluate the delay-based detector proposed in [10] as our first detector design because it is less complex to implement than other types.

979-8-3503-4135-5/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: University of Grenoble Alpes. Downloaded on November 02,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

This detector was synthesized using a VHDL RTL description in the Xilinx Vivado design suite. We determined that the threshold detection frequency of this detector is 17.2 MHz.

As many detectors rely on Ring Oscillators (RO), the second Device Under Test (DUT) is a basic RO design. Furthermore, compared to delay-based detectors, RO-based detectors offer a better understanding of how power-off attacks function. To conduct our experiment, we will focus on three main scenarios. The first scenario is power-off, where DUTs are turned off and not connected to any source of power. In the second scenario, called power-on, both DUTs are on. This scenario helps us compare the results of this scenario with the power-off scenario. And the last scenario is clock freezing, which means the detector is on but its clock is frozen (i.e., does not have any edge), and similarly for RO, the RO enable signal is zeroed (i.e., the RO is inactive).

III. Experiments and Results

In this section, we present experiments conducted on actual devices to evaluate the impact of heating on the scenarios described in the previous section.

As mentioned in the previous sections, our objective is to evaluate the impact of overheating on the performance of the delay-based detector and the simple Ring Oscillator (RO) when the power is off. For each DUT, we examined the three previous scenarios. We implemented each scenario on a separate BASYS-3 board, which contains a Xilinx Artix7 chip. To conduct the practical evaluation, we utilized the Votsch VC 0018 climate test system chamber, capable of producing heat up to +95°C. We subjected the delay-based detector and RO to various time cycles within the thermal chamber to examine the impact of heat on their performance. To evaluate the detector's response time, we utilized a simple LED to measure the alarm signal's activation time, while for the RO, we used a frequency meter. All measurements were taken with the chips held at the same temperature, namely normal room temperature.

Table I and Table II display the results of the test conducted for a period of one week, in these two tables, the variations in activation thresholds of the two preceding detectors are represented. These variations are expressed as a percentage and are given after 1 day, 2 days, and 7 days in the climatic chamber. The 3 previously presented scenarios are used (power-on, clock-freezing, and power-off). 3 FPGAs are used for each type of detector in these 3 scenarios. This temperature is the maximum temperature that the climatic chamber can deliver. It is lower than the temperature that a laser could locally achieve. However, in the context of this study, we want to see if we observe a temperature attack effect when the circuit's power is off. If we observe an effect by immersing the entire circuit in the climatic chamber, a laser could certainly create the same variations, with the difference that a laser with high precision in the location of the fault injection and also with a higher temperature can achieve this effect in a short period of time compared to our work.

TABLE I
Results of a heating delay based detector

Percent change of Alarm activation (%)			
Time under test	power-on	clock-freezing	power-off
Initial value	0	0	0
After 1-day	0	0	-0.3
After 2-day	-1.7	-3.2	-0.2
After 7-day	-6.4	-4.2	-1.3

TABLE II
Results of a heating RO

Percent change of Ring Oscillator(RO) frequency (%)			
Time under test	power-on (en=1)	clock-freezing (en=0)	power-off
Initial value	0	0	0
After 1-day	0.2	-0.5	0
After 2-day	-1.0	-0.5	0
After 7-day	-1.0	-0.9	-0.7

IV. DISCUSSION

Heating cycles are effective in reducing both the delay-based detector's threshold detection frequency and the ring oscillator's frequency for all the scenarios including the power-off mode. But, for all scenarios, the ring oscillator design is less affected than the delay-based detector. For instance, in power-on mode after seven days, the threshold detection decrease is 6.4% rather than the ring oscillator frequency decrease is 1%. In power-off mode after seven days, the threshold detection decrease is only 1.3% rather than the ring oscillator frequency decrease is 0.7%. Finally, the clock-freezing mode has a lesser effect than the power-on mode and a higher effect than the power-off mode. We believe that, in clock-freezing mode, as transistors remain stable (i.e., permanently open or close), this implies a Negative Bias Temperature Instability (NBTI) [11] affecting only a part of the PMOS transistors (the always opened transistors). As the effect of NBTI is non-linear with the duration, this makes the aging effects lesser in this mode than in the power-on state (with a living clock) where all the PMOS transistors are then impacted by NBTI. Note that we do not know which effect is observed during the power-off mode where NBTI cannot be responsible as no power is applied. As far as we know, no paper has described the effect of heating on power-off circuits.

V. CONCLUSION AND PERSPECTIVES

The objective of our study was to evaluate the effects of power-off attacks on digital circuits and the susceptibility of delay-based detectors. These types of attacks can jeopardize chip security since POAs cannot be detected by active mechanisms. In particular, we will look for detectors impacted by POAs to create false negatives. More specifically, within the context of the POP project (mentioned in the acknowledgment section), our plan is to use a laser fault injection technique with high fault locality and produce a dedicated test chip to conduct comparable experiments. These experiments will involve localized temperature attacks on various digital and analog circuits, including detectors.

Authorized licensed use limited to: University of Grenoble Alpes. Downloaded on November 02,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

VI. Acknowledgement

This work was supported by a research grant from the French Agence Nationale de la Recherche (POP project, ANR-21-CE39-0004). [12]

References

[1] P. Kocher, J. Jaffe, and B. Jun, "Differential power analysis," in *Annual international cryptology conference*. Springer, 1999, pp. 388–397.

[2] D. Boneh, R. A. DeMillo, and R. J. Lipton, "On the importance of checking cryptographic protocols for faults," in *International conference on the theory and applications of cryptographic techniques*. Springer, 1997, pp. 37–51.

[3] J. Breier, S. Bhasin, and W. He, "An electromagnetic fault injection sensor using hogge phase-detector," in *2017 18th International Symposium on Quality Electronic Design (ISQED)*, 2017, pp. 307–312.

[4] M. R. Muttaki, T. Zhang, M. Tehranipoor, and F. Farahmandi, "Ftc: A universal sensor for fault injection attack detection," in *2022 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, 2022, pp. 117–120.

[5] D. El-Baze, J.-B. Rigaud, and P. Maurine, "An embedded digital sensor against em and bb fault injection," in *2016 Workshop on Fault Diagnosis and Tolerance in Cryptography (FDTC)*. IEEE, 2016, pp. 78–86.

[6] H. Igarashi, Y. Shi, M. Yanagisawa, and N. Togawa, "Concurrent faulty clock detection for crypto circuits against clock glitch based dfa," in *2013 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2013, pp. 1432–1435.

[7] A. G. Yanci, S. Pickles, and T. Arslan, "Characterization of a voltage glitch attack detector for secure devices," in *2009 Symposium on Bio-inspired Learning and Intelligent Systems for Security*. IEEE, 2009, pp. 91–96.

[8] J. Richter-Brockmann, A. R. Shahmirzadi, P. Sasdrich, A. Moradi, and T. Güneysu, "Fiver–robust verification of countermeasures against fault injections," *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pp. 447–473, 2021.

[9] N. Selmane, S. Bhasin, S. Guilley, and J.-L. Danger, "Security evaluation of application-specific integrated circuits and field programmable gate arrays against setup time violation attacks," *IET information security*, vol. 5, no. 4, pp. 181–190, 2011.

[10] L. Zussa, A. Dehbaoui, K. Tobich, J.-M. Dutertre, P. Maurine, L. Guillaume-Sage, J. Clediere, and A. Tria, "Efficiency of a glitch detector against electromagnetic fault injection," in *2014 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2014, pp. 1–6.

[11] Z. Ghaderi, *Aging-induced Performance Degradation: Monitoring and Mitigation*. University of California, Irvine, 2017.

[12] V. Beroulle, B. colombie, G. Natale, P. Maistri, and I. Vatajelu, "Design and evaluation of countermeasures against power-o laser fault injection attacks," https://tima.univ-grenoble-alpes.fr/sites/tima/files/Mediatheque/Jobs/PhD/sujet$_t hese_p op.pdf$, $May 2022$.

Authorized licensed use limited to: University of Grenoble Alpes. Downloaded on November 02,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.